The Ephemeris

March 2017

Volume 28 Number 01 - The Official Publication of the San Jose Astronomical Association

March - June 2017 Events

Board & General Meetings
Saturday 3/11, 4/8, 5/6, 6/10
Board Meetings: 6 -7:30pm
General Meetings: 7:30-9:30pm

Fix-It Day (2-4pm)
Sunday 3/5, 4/2, 5/7, 6/4

Solar Observing (locations differ)
Sunday 3/5, 4/2, 5/7, 6/4

Intro to the Night Sky Class
Houge Park 1Q In-Town Star Party
Friday 3/3, 3/31, 4/28, 6/2

Astronomy 101 Class
Houge Park 3Q In Town Star Party
Friday 3/17, 4/14, 5/19, 6/16

RCDO Starry Nights Star Party
Saturday 3/18, 4/15, 5/20, 6/17

Imaging SIG Mtg
Tuesday 3/21, 4/18, 5/16, 6/20

Astro Imaging Workshops
(at Houge Park)
Saturday 3/25, 6/24, 9/23, 12/16

Coder's Meeting
Sunday 3/19, 4/23, 5/21, 6/11

Quick STARt (by appointment)
Friday 4/7, 6/9, 8/12

Binocular Star Gazing
Saturday 5/20, 6/17, 7/15, 8/12

Unless noted above, please refer to the SJAA Web page for specific event times, locations and possible cancelation due to weather.

INSIDE THIS ISSUE

Cover Photo: Andromeda Galaxy

By SJAA Member: Eric Zbinden

SJAA Annual Meeting

Pot Luck Dinner, Elections and Awards

February 11, 2017

From Tom Piller

Board members Dave Ittner, Ed Wong, Rob Chapman, Teruo Utsumi, Wolf Witt, Glenn Newell, Bill O'Neil, Rob Jaworski and Vini Carter adjorn the February 2017 board meeting. Lee Hoglan not pictured.

Carter, Chock and Jaworski unveil the new SJAA banner

This year at the February11th annual meeting, food was served, elections were held and members were recognized.

Dave Ittner did an outstanding job of noting all that our club does and handed out service recognition awards to Ed Wong and Lee Hoglan for their contributions to SJAA's various community outreach programs. At the end of the awards, Dave was surprised by Director Bill O'Neil with an award of his own for his two years of service as SJAA's President.

SJAA school astronomy and star party program, chaired by Jim Van Nuland, was also recognized by Dave as one of the more important outreach programs provided by SJAA for the hundreds of children and teens it reaches at schools in the South Bay Area.

The elected board members for the five open seats in 2017 were Sukhada Palav, Vini Carter, Robert Chapman, Swami Nigam and Teruo Utsumi.

Following below and on the next page are a few more pictures of the evening. Photo credits go to Ed Wong and Tom Piller.

SJAA Annual Meeting

Pot Luck Dinner, Elections and Awards

Immediately above:

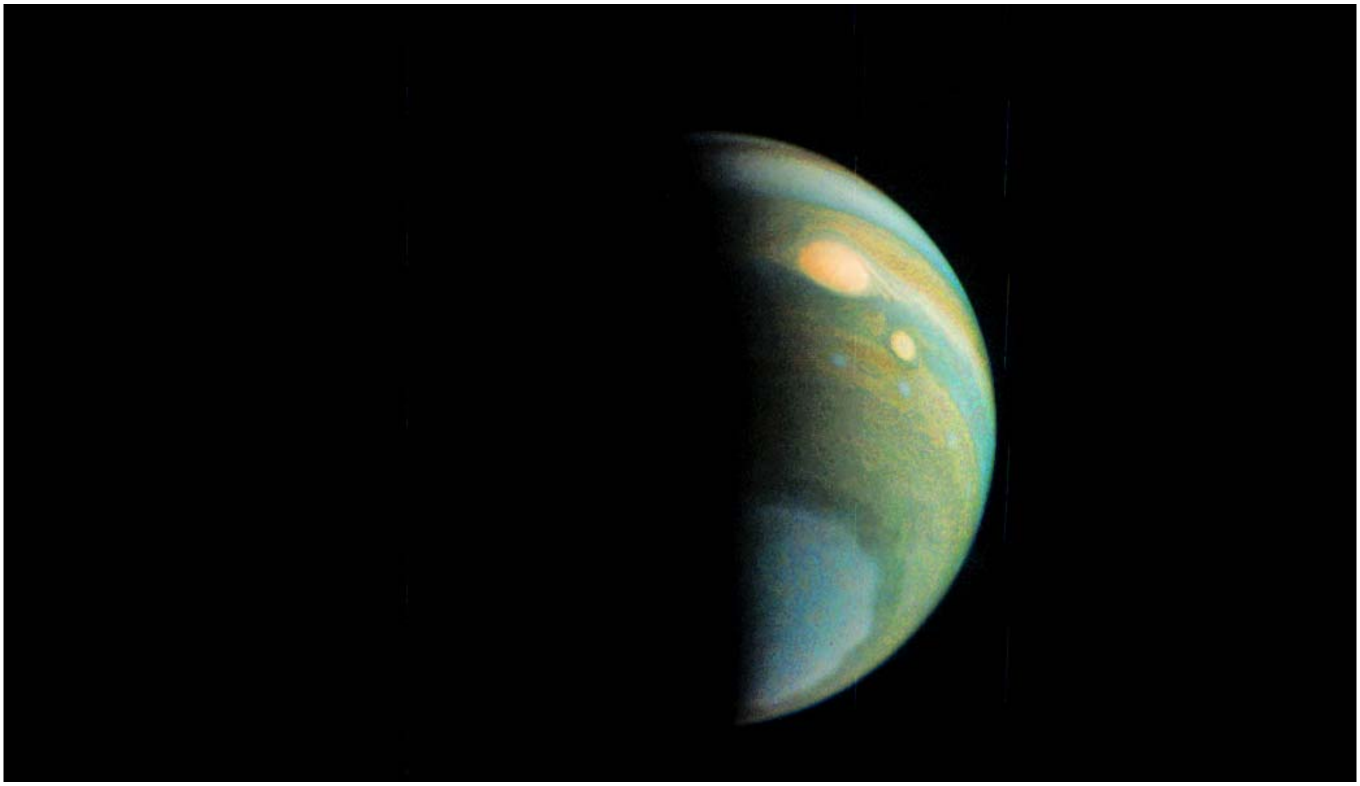
Ed Wong receives an achievement award in recognition of his service to SJAA Programs most notably Binocular Stargazing and Pinnacles National Park Star Parties.

Right: Outgoing Pres. Dave Ittner receives a surprise award from Director Bill O'Neil for his tireless service as President of SJAA for the last two years.

Immediately above:

Rob Jaworski receives a service achievement award for Lee Hoglan (not pictured) for his many years of service in the SJAA Club. One of Lee's most notable recent contributions were the years he spent as "Ask Lee" fielding questions from the public on the SJAA web page. Lee has been a board member since March 2006.

Never Groundhog Day in Jupiter



This false color view of Jupiter's polar haze was created by citizen scientist Gerald Eichstädt using data from the JunoCam instrument on NASA's Juno spacecraft.

Credits: NASA/JPL-Caltech/SwRI/MSSS/Gerald Eichstädt

NASA's Juno mission completed a close flyby of Jupiter on Thursday, Feb. 2, its latest science orbit of the mission.

All of Juno's science instruments and the spacecraft's Juno Cam were operating during the flyby to collect data that is now being returned to Earth. Juno is currently in a 53-day orbit, and its next close flyby of Jupiter will occur on March 27, 2017.

At the time of closest approach (called perijove), Juno will be about 2,670 miles (4,300 kilometers) above the planet's cloud tops and traveling at a speed of about 129,000 mph (57.8 kilometers per second) relative to the gas giant. All of Juno's eight science instruments, including the Jovian Infrared Auroral Mapper (JIRAM) instrument, will be on and collecting data during the flyby.

"Tomorrow may be 'Groundhog Day' here on Earth, but it's never Groundhog Day when you are flying past Jupiter," said Scott Bolton, principal investigator of Juno from the Southwest Research Institute in San Antonio. "With every close flyby we are finding something new."

The Juno science team continues to analyze returns from previous flybys. Revelations include that Jupiter's magnetic fields and aurora are bigger and more powerful than originally thought and that the belts and zones that give the gas giant's cloud top its distinctive look extend deep into the planet's interior. Peer-reviewed papers with more in-depth science results from Juno's first three flybys are expected to be published within the next few months. Also, JunoCam, the first interplanetary outreach camera, is now being guided with the assistance from the public -- people can participate by voting for what features on Jupiter should be imaged during each flyby.

Editor's note: If you are interested in voting on and possibly influencing what JunoCam takes pictures of the site link is listed beloiw:

https://www.missionjuno.swri.edu/junocam

SJAA Member Observation & Trip Report Posts

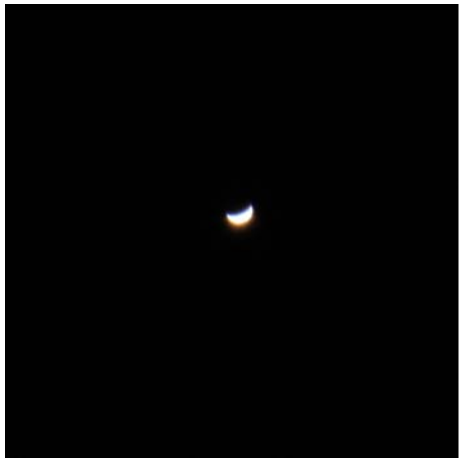
Observing planets Neptune, Uranus, and Venus

From Marilyn Perry

On New Year's Eve, I observed Neptune when it was only 5 arcmin from Mars in the Southwest sky. The two planets easily fit in the same field of view of my high power eyepiece making it very easy to find Neptune. I really had to use my imagination to see that Neptune at magnitude 7.9 had the features of a planet: extremely subtle blue color and lack of twinkling. What surprised me the most was that I was seeing Neptune in nearly the same area of the sky where I had plotted it for the Astronomical League's (AL) Galileo program over a year ago. Neptune had only moved a little over 2 degrees in about a year. Neptune has an orbital period of about 165 earth years, so that would be about right.

The previous night I had observed Uranus for the AL's Solar System program. I picked a wonderful night because three stars of about the same magnitude as Uranus at magnitude 5.8 fit in the same low power eyepiece field of view. Transparency was not the best, and seeing was poor. The stars were twinkling like mad, but there was not a hint of twinkling for planet Uranus. The stars looked like points of light, but Uranus looked like a subtle blue disk, a very definite contrast between stars and planet.

For AL's Galileo program and now for AL's Solar System program, I am observing Venus on a regular basis from one inferior conjunction, 8/15/2015 to the next, 3/25/2017. Inferior conjunction is when Venus lies between the earth and the sun, hence is nearest the earth. Venus takes the same crescent and gibbous shapes as does the moon. But whereas the moon spends as much time appearing in crescent shape as it does in gibbous shape, Venus spends much more time in gibbous shape than in crescent shape, about 3/4 of the time. It just switched from gibbous to crescent around greatest elongation, 1/12/2017. To me the crescent shape is more visually appealing. The moon is always about the same distance from the earth so it is much brighter when in gibbous shape. By contrast Venus is closer to the earth when in crescent shape, so it is bigger and brighter then. It will be brightest on February 18 when it is roughly one quarter illuminated.



Because of clouds, I was not able to see Venus when it was really slender back in 2015. I am hoping to do better this year. I read in the March issue of Sky and Telescope magazine that for the 2017 inferior conjunction, Venus will pass more than 8 degrees north of the sun, and for several days before or after 3/25/2017 some observers at our latitude will be able to see the very thin crescent Venus both in the morning and evening. With my high horizons and with all the clouds we have been having, I doubt that I will be able to do this, but I will do my best to see the most slender Venus I can. I invite you to also enjoy the very slender Venus. It is not a slender crescent for long, and now is our short chance to see it. If we miss this chance, the next inferior conjunction will occur October 2018.

Williams Hill Site Report

From Rob Jaworski

Editor's Note: The following is an excerpt from Rob Jaworski's original report which can be found on the SJAA web page under the Trips section of the SJAA Blog….Piller

This is a site report, not so much an observing report. My main goal was to go to check out the surroundings. On July 30-31 2016, I went to Williams Hill Recreation Area, managed by the Bureau of Land Management (BLM), to investigate the suitability of the site for purposes of astronomical observing, with the possibility of spending the night. The area is located about 27 road miles, or 17 miles as the crow flies, south of King City.

My plan was to get there earlier in the day on Saturday with my dirt bike, and then tool around to explore and get to know the area, and settling on a location where I could set up my scope. I would then have dinner, and head to that site to set up, and wait for night to fall. Things went very much according to plan.

The ride down 101 was uneventful. It was over a hundred miles from San Jose on southbound 101, through all the towns we know when we are traveling to dark sky sites: Gonzales, Soledad, Greenfield, King City.

The exit off of 101 to take is called San Ardo, followed by a right onto Paris Valley Road. After a little less than two miles of paved farm road, you have to make a sharp left turn onto Lockwood San Ardo Road, at which point, it turns to a fairly well maintained dirt road. Once on the dirt, going was slow. Nearing the camp, the road is cut into the side of the hill, exposing the Monterey shale formations. It's a very dry, dusty white ride. The dirt road leading to Williams Hill is about eight miles of white dusty dirt road.

Once you see this sign, you know you've made it!

continued from page 4…..Williams Hill Report

I arrived at the campground closer to 4pm, about ninety minutes later than I originally planned. The first order of business was to unhitch the trailer and pull the bike down, getting ready for the first phase, local exploration.

The campground is four and a half miles before the end of the road that leads up to the actual Williams Hill, which can also rightly be called Antenna Hill. I made my way to antenna hill via the fairly well maintained dirt road comprised mainly of switchbacks that lead from one hill to the next. At the top of the hill, the antennae live behind a chain link fence, some of the towers more securely enclosed than others, possibly giving away their importance to their owners. At the end of the road, there is a fairly wide area that may be suitable to set up maybe a dozen scopes, but it's a long way to go past the campground, there are large circles etched into the ground by spinning wheels (ie, donuts), and if anyone were to come up that road at night, you'd be blasted with headlights with little to no cover. Further, the flat area, though wide open at the top of the hill with nice horizons (except for the towers), is not entirely level. Being the top of a hill, it has a mild crown, I thought to myself that this area would do in a pinch, but there very well could be better sites.

The actual Williams Hill, where all the antenna towers are. It's four and a half miles past the campground.

I kept exploring and found miles of more trails on this BLM land. North of the campground area, ie, north of Lockwood San Ardo Road, there are plenty of possible observing sites with decent horizons and plenty of space for plenty of astronomers' vehicles and their gear. Many, most or all of them were littered with spent shotgun shells. It was shameful, how the shooters left such a mess. There weren't too many targets, and thankfully not much broken glass, though a gallon size fire extinguisher appeared to be on the receiving end of a muzzle.

After a good hour and a half of exploring, I concluded that the best overall site for observing is only about half mile from the campground, at the following coordinates:

35.984279, -121.011742.

It's an open hill good with horizons nearly all the way around, especially to the east. So much so that you can look down and see headlights on highway 101. After dark, there was just a small light dome to the east, this is most likely skyglow.

The view of the road leading up to Williams Hill Recreation Area, from the top, looking east.

The campground is wooded with pines, though sparsely, and each of the six sites are far enough apart which provides for privacy. The location is on top of a ridge, so the wind was sailing briskly over the ridge top, making tent assembly a bit difficulty. However, it died down rather abruptly after about 7pm.

The campsite showing fire ring, canopy, picnic table.

There is no water at the campground whatsoever, so any campers or other visitors are well advised to bring plenty of their own. There is a single vault toilet that was fairly (relatively) clean and even stocked with adequate amounts of TP. The sites each had a metal picnic table and a canopy as shelter from the sun, or perhaps the rain, and the sites and much of the rest of the campground were roped off to keep off roaders off the vegetation. There were other campers, including a fairly large group of young men, in about 6 cars. They were not any problem, we didn't socialize at all, and the smell of clove cigarettes adorned the air on the road leading away from camp. There was also a later arrival, a guy with his girlfriend, out for a romantic camp getaway in the local hills.

Rob Jaworski's tent and campsite at sunset

As far as observing, I don't have much to write about. The hilltop location were I set up was mildly rutted by vehicles turning around when the ground was last wet. The ruts made it slightly challenging to find an adequately flat, level spot for the dob base. It wasn't dusty there, but rather a bit rocky with small scrub growing low. After it got dark, I sampled Saturn, whose tilt is now so great that seeing the Cassini division is easy, and enjoyed the planets wandering through Scorpius. viewed some objects around Ophiucus but mainly I was happy just taking in the dark sky, with the milky way hanging above like fluffy clouds. A few streaks of light appeared during the night, perhaps a pregame show for the coming Perseid meteor shower. It was very much plenty dark.

I didn't stay out long since I had to be out of there early. The next morning, I loaded up and took it easy down the hill again, and it was a nicer ride since it wasn't as hot. Got back to San Jose around 11am.

Would I recommend this place for SJAA members to go observing? Definitely yes. The recommendation would also include that you should plan on staying the night. Though there is no water and no trash service, and it's a fairly remote location where shooting is allowed, it does have fairly decent cell service, a nice campground, and good horizons at 2200 feet (671 meters).

SJAA Member Observation & Program Posts

Venus and 1 day old Moon

Email string originated with Dave Ittner

Message 1 of 3 , Dec 1, 2016
I went outside 30 minutes ago and set up the AR127 to get ready for some backyard observing tonight.
Started up Stellarium and noticed that there was a 1 day old moon (http://www.moongiant.com/phase/today) and Venus.
I spotted both naked eye then looked at them through the scope as well as my 20x80 binos.

Reply Ed Wong
Dec 1, 2016
I held a star party for the students at my school tonight and we also had a chance to see both [Venus and the Moon]. The students want to start a astronomy club on campus and now they want me to help them.

Reply Marilyn Perry
Message 3 of 3 , Dec 2, 2016
I showed the moon to early-arrivers at the SJAA school party last night, and we all agreed it was beautiful. There were three big attractive craters in a vertical line that I have never paid any attention to: top to bottom, Langrenus, Vendelinus, and Petavius. They are about the same size, but they made a nice contrast because the one in the middle has a smooth floor whereas the two on the ends have very easily-seen peaks. The adults in the know were commenting on how well they could make out features in the earthshine.

Moon will soon occult Aldebaran

(Occult occurred on December 12, 2016)

Excerpts from email string originated with Marilyn Perry
Message 1 of 6 , Dec 12, 2016
According to Sky Safari, the moon will occult Aldebaran from around 7:06 PM until around 8:02 tonight. Right now there are no clouds near the moon and Aldebaran easily fits in the same field of view in my binoculars. There have been so many clouds today that I don't trust that the sky around the moon will stay clear, but it might. It will be a more impressive sight than when I viewed the moon occulting a dimmer star a couple of months ago.

Marilyn Perry
Message 4 of 6 , Dec 12, 2016
That was so fun! I was astonished when Aldebaran winked out in an instant. Looking at Sky Safari and also looking at the moon, the moon was so round that I didn't get the idea that the edge near Aldebaran was the terminator, not the limb. I was listening to my talking clock and thinking that it was supposed to occult in less than a minute, but there still was quite a bit of space between Aldebaran and the lighted moon, and then in an instant Aldebaran winked out. Then I figured out that it went behind the dark area of the moon. So different than when 89 Tau was occulted by the limb as it eased into the bright glow around the lighted limb and then was gradually lost in the glow.

It was worth my challenges with the clouds. The area around the moon was all clear when I decided to set up my grab and go telescope, but by the time I was all set up the clouds had moved in. They were in stripes, I guess you call them mackerel scales. As I watched the moon it would be alternately bright and dim to very dim.

Aldebaran is very bright; it punched through all but the thickest of the clouds. It fit in the same 142x FOV with a crater with very high albedo, Aristarchus. Even when looking through the thickest of the clouds, I could still see Aristarchus, so I didn't get lost. Talking clock said 7:05, a thick cloud rolled by, I just barely saw the bright crater, a break in the cloud rolled by, I saw Aldebaran very clearly, in an instant it was gone, I was amazed, the talking clock said 7:06. So fun!!!

2016-2017 School events as of Feb.11, 2017

From Jim Van Nuland

Completed school events as of Aug '16 - Feb. 11, 2017

	Total sched	Good sky	Partial success	Cloudy failed	Cancelled at noon
Aug	2	2	0	0	0
Sep	0	0	0	0	0
Oct	4	2	0	0	2
Nov	8	5	0	0	3
Dec	5	0	4	0	1
Jan	5	1	0	0	4
Feb	4	1	0	0	3 (to Feb.10)
Total	28	11	4	0	13

Scheduled events as of Dec. 8, 2016

	Total	Firm	Working
Feb	1	1	0
Mar	5	4	1
Apr	1	0	1
May	3	2	1
Tot al	10	7	3

"Working" means that a date has been chosen, but not all approvals have been obtained. With a firm date, some details may remain to be worked out.

Clear Skies!
Jim Van Nuland, Chairman, School Star Party program

SJAA Member Observation Posts

Mendoza Ranch
Saturday 1/28/2017

From Gary Chock

It was great to share the views with everyone at Mendoza Ranch. Good to be outside after all the Atmospheric Rivers. The temperature started off pleasant, but got quite chilly due to the wind. It was clear, except for some patchy high cirrus every so often. Following is a list of some of the objects we viewed.

- M42 Orion Nebula, nice nebulosity, could only see the 4 Trapezium stars A-B-C-D, no E
- Sigma Ori, as Marilyn and Vini suggested, could not split AB, could see D, E, C. Noticed that D was out of line from the chart that Vini found. (Sorry I shouted out the incorrect Greek letter, Sukhada! it's Sigma, not Gamma!)
- Iota Ori, could not split; forgot about Beta Mon!
- NGC 404 Mirach's Ghost, as Chis suggested
- M31, M32, M110, Andromeda Galaxy et al, nice view through binoculars
- M33 Triangulum Galaxy, nice patches and clumps
- NGC 891, edge-on spiral galaxy, maybe discern dust lane with averted "imagination"
- M81 and M82, Bode's and Cigar Galaxies
- M65, M66, NGC 3626 Leo Triplet
- M95, M96 second Leo Triplet, could not make out M105
- NGC 1023, as Chis suggested
- M97 Owl Nebula
- M51 Whirlpool Galaxy, a little hazy to the north above horizon
- NGC 2392 Eskimo Nebula, averted vision helps to bring out nebula and star
- M46 and M47, NGC 2438 planetary nebula in M46
- NGC 2239 Rosette Nebula, narrowband filter and panning the FOV helped bring out the expanding nebula
- M1 Crab Nebula, narrowband and OIII filters brought out great structure, "veins"
- NGC 869 and NGC 884 Double Cluster, beautiful view through Vini's refractor, noted the string of stars leading to Stock 2

Astroimaging Report
December 2016

From Glenn Newell

December 2016's activity was a combined workshop and field clinic at Coyote Valley with 32 people in attendance. https://www.meetup.com/SJ-Astronomy/events/225111455/ 8 Photos were uploaded to the following address:

https://www.meetup.com/SJ-Astronomy/photos/27496303/#456960936

This was the first time giving a formal lecture with slides (outdoor projector and screen) on all the different types of astrophotography, while waiting for Polaris to be visible. The program seemed to be a hit as you can see from feedback on the SJAA meetup page.

We also successfully captured some images with an attendee's camera on the club DSLR rig and was able to send him home with some data to process. Bruce Braunstein was able to help a photographer just getting started with nightscapes, so many thanks to Bruce for his efforts.

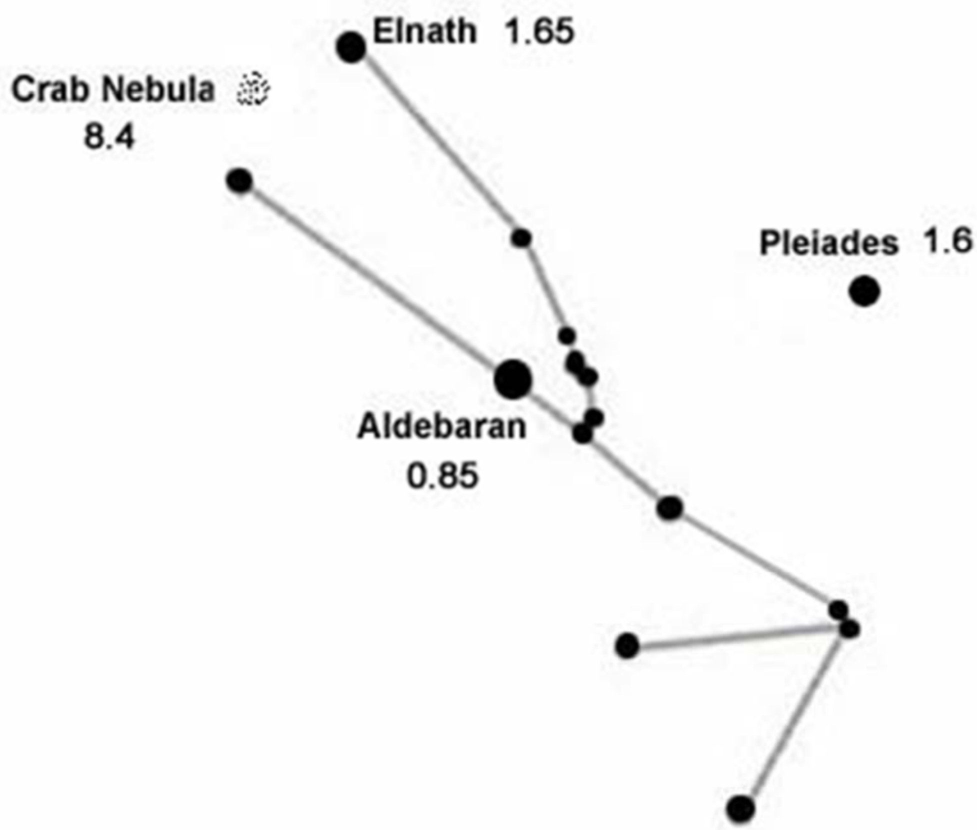
In addition to deep sky with SJAA's DSLR rig, I was also able to demonstrate polar alignment with the Club's iOpteron SkyTracker.

So all in all a successful event.

Below is some LRGB data of the Flame Nebula: and narrowband data on the Medusa Nebula:

Aldebaran

From Sandy Mohan



Aldebaran, designated Alpha Tauri (α Tauri, abbreviated Alpha Tau, α Tau), is an orange giant star located about 65 light years from the sun in the zodiac constellation of Taurus. It is the brightest star in its constellation and usually the fourteenth-brightest star in the nighttime sky, though it varies slowly in brightness between magnitude 0.75 and 0.95. It is likely that Aldebaran hosts a planet several times the size of Jupiter.

The planetary exploration probe Pioneer 10 is currently heading in the general direction of the star and should make its closest approach in about two million years.

Aldebaran is one of the easiest stars to find in the night sky, partly due to its brightness and partly due to its spatial relation to one of the more noticeable asterisms in the sky. If one follows the three stars of Orion's belt from left to right (in the Northern Hemisphere) or right to left (in the Southern), the first bright star found by continuing that line is Aldebaran.

Since the star is located (by chance) in the line of sight between the Earth and the Hyades, it has the appearance of being the brightest member of the more scattered Hyades open star cluster that makes up the bull's-head-shaped asterism; however, the star cluster is actually more than twice as far away, at about 150 light years.

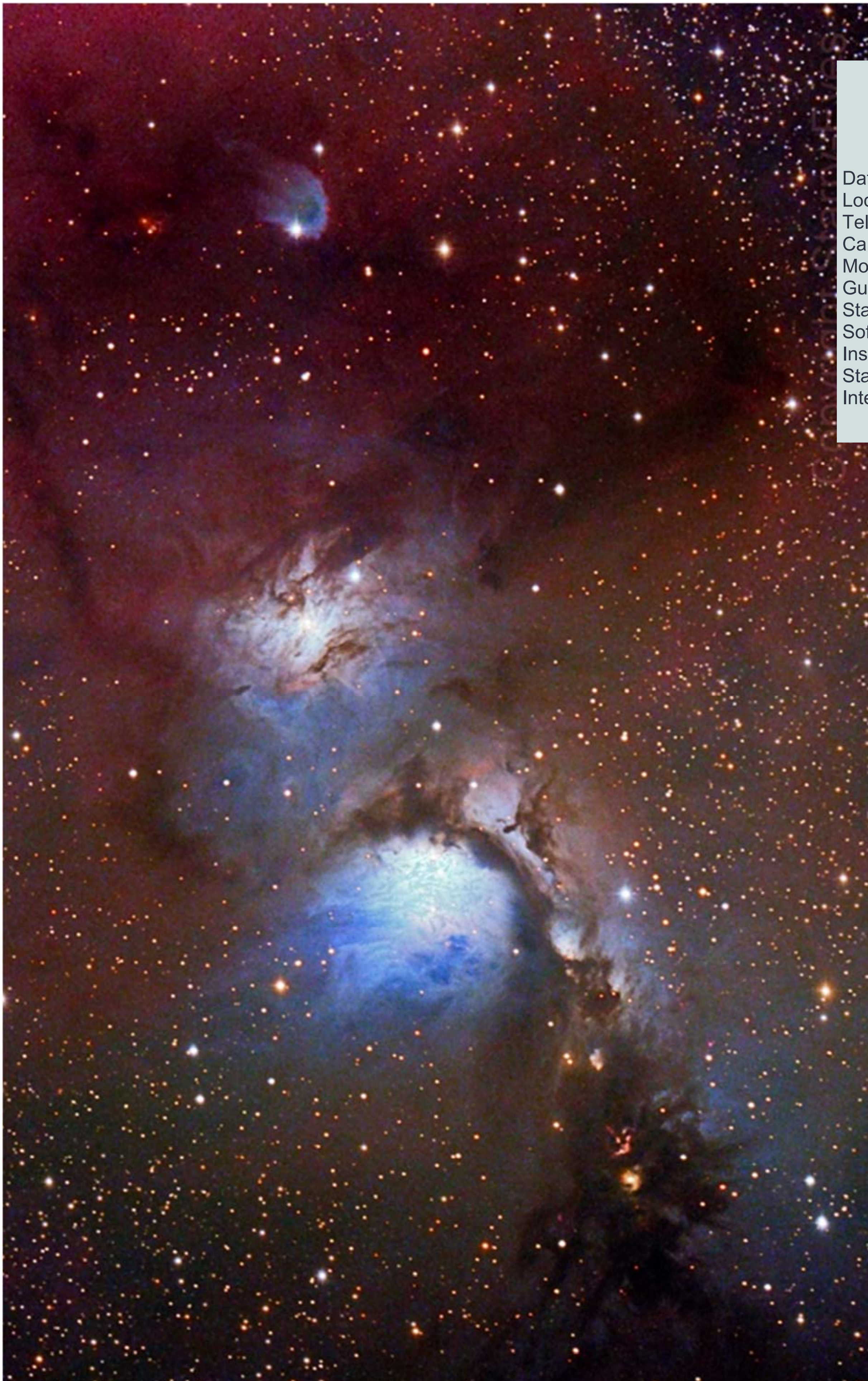
Aldebaran is close enough to the ecliptic to be occulted by the Moon. Such occultations occur when the Moon's ascending node is near the autumnal equinox. A series of 49 occultations occur starting at 29 Jan 2015 and ending at 3 Sep 2018. Each event is visible from a different location on Earth, but always in the northern hemisphere or close to the equator. That means that people in e.g. Australia or South Africa can never observe an Aldebaran occultation. This is due to the fact that Aldebaran is slightly too far south of the ecliptic. A reasonably accurate estimate for the diameter of Aldebaran was obtained during the September 22, 1978 occultation. Aldebaran is in conjunction with the Sun around June 1 of each year. Source: Wikipedia

SJAA Member Astrophoto Gallery

M31 and Mystery Ha Clouds
From: Rogelio Bernal Andreo

Location: DARC Observatory, Henry Coe State Park, Montebello OSP
OTA: Dual set up of two 4" Takahashi FSQ106EDX
Camera: SBIG STL 11000, roughly 47h of Ha and 15h of LRGB
Stacking program: PixInsight
Processing program: PixInsight, Photoshop
See Also: http://www.deepskycolors.com/Clouds_of_Andromeda.html

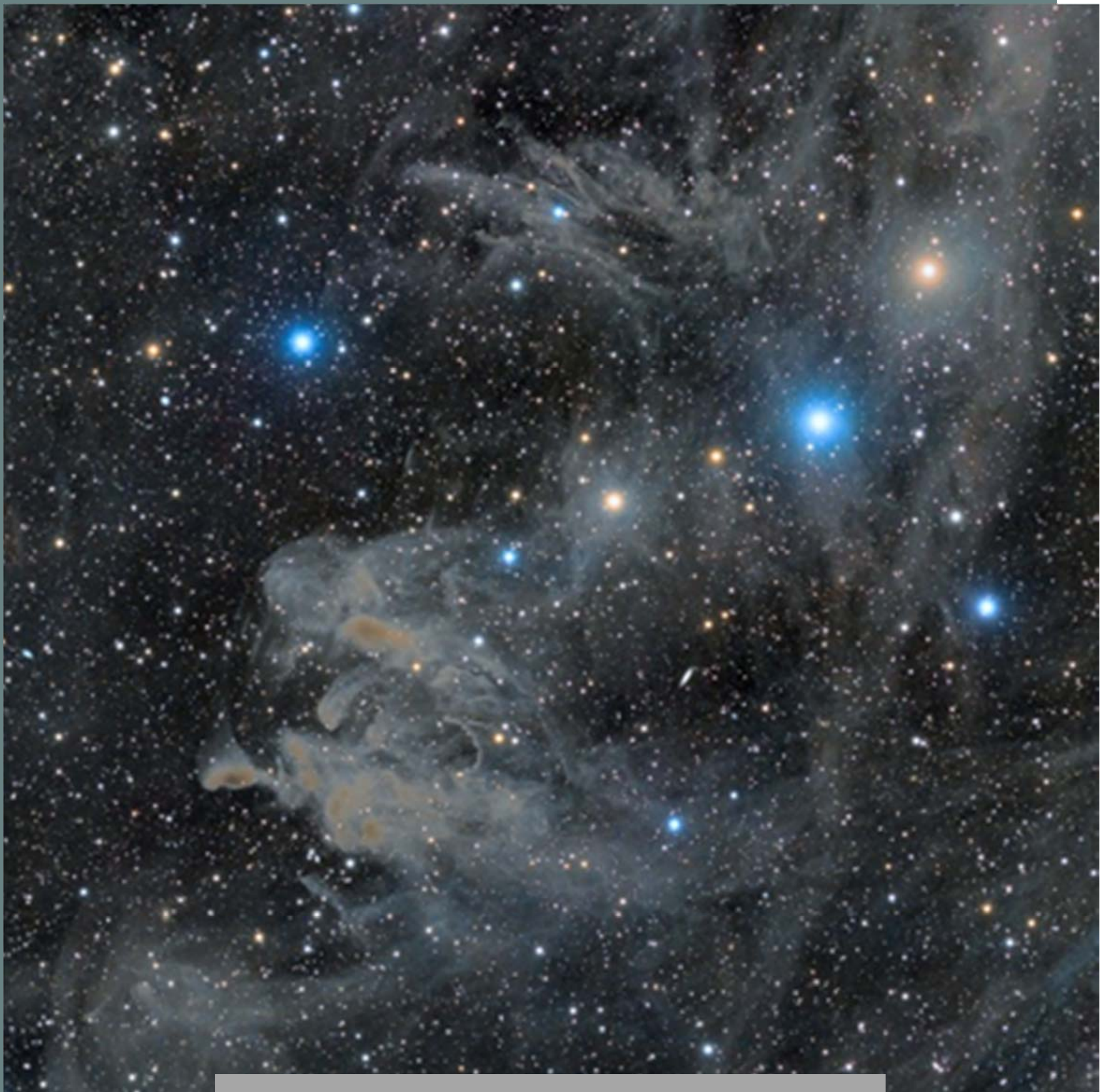
Rogelio was born in Spain but has been living in the United States for almost 30 years. He commenced producing astronomical photographs nine years ago. His work has been featured on APOD 47 times, published in several astronomy publications, used in planetariums, astronomy exhibits at museums, and appeared in the IMAX/Warner Bros. motion picture production Hubble 3D. Rather than simply trying to obtain the *best* image, he constantly challenges himself to ensure the final picture connects with the viewer by focusing on composition and experimenting with new processing techniques. Rogelio has spoken for the SJAA on several occasions and, although he does most his astrophotography from a dark private location, he is an active "nomadic" imager across many of the South Bay dark sites.



M 78

From PJ Mahany

Date: Jan. 16, 2017
Location: Yosemite, California
Telescope: Celestron RASA
Camera: Canon 60Da
Mount: Orion HDX110 EQ8
Guiding: Orion 50mm with Starshoot Autoguider
Software: SkyTools, Pix-Insight1.8, BYEOS, PHD 2, StarTools
Integration: 2.7 hours



LBN 762 and LBN 753 Molecular Clouds in Aries

From Eric Zbinden

Location: Lake San Antonio, CA (CalStar 2010)
Date: September 2010
Scope: AP155 F5.3
Mount: AP900GTO
Camera: FLI PL16803
Exposure: LRGB: 600:120:120:120

Kid Spot Jokes:

- **Why didn't the Dog Star laugh at the joke?**
  It was too Sirius.
- **How does the Solar system hold up its pants?**
  With the Asteroid Belt.
- **Why didn't the astronaut get burned when he visited the Sun?**
  He went at night.

M31
Photo: Eric Zbinden

Kid Spot Quiz:

1. **Which organization is responsible for naming constellations?**
2. **How many constellations have been named?**
3. **What was the name of Buzz Aldrin's (second person to walk on the moon) mother?**

100 Years of Astronomical Discovery

Excerpts From Astronomy Magazine September 2016 Edition

1904 German astronomer Johannes Franz Hartman discovers the interstellar medium.

1905 Albert Einstein describes the photoelectric effect, introducing the concept of photons.

1905 Einstein publishes "On the Electrodynamics of Moving Bodies" in which he outlines special relativity.

1906 German astronomer Max Wolf discovers 588 Achilles the first identified Trojan asteroid.

1908 Something, likely a small asteroid or comet, levels some 80 million trees along the Stony Tunguska River in Russia.

1908 The 60" Hale telescope sees first light atop Mount Wilson, California. It was at the time the largest operational telescope in the world.

1908 American astronomer George Ellery Hale discovers magnetic fields in sunspots.

1909 Swedish astronomer Karl Bohlin proposes the idea that the Sun is not in the center of the Milky Way.

1912 American astronomer Vesto Slipher takes the first spectrogram of the **Andromeda Nebula** (M31). It reveals a Doppler shift, the first such recorded.

Kid Spot Quiz Answers:

1. **International Astronomical Union**
2. **88**
3. **Marion Moon**

From the Board of Directors

Announcements

- ◊ SJAA Club Officer elections will be held at the 3/11/2017 meeting.
- ◊ The Spring 2017 Swap Meet will be held on Sunday, 3/12/2017
- ◊ Cupertino Earth Day, scheduled on April 22, 2017

Board Meeting Excerpts

November 12, 2016

In attendance

Dave Ittner, Rob Chapman, Rob Jaworski, Bill O'Neil, Lee Hoglan, Vini Carter, Glenn Newell, Teruo Utsumi
Absent: Ed Wong (Excused)
Guests: Michael Phan, Swami N, Anand Rajagopalan, Gary Chock, Ranjun Desai

SJAA Membership Database

Swami provided a proposal to create SJAA Membership Database which could be accessed through a web interface. Auto-renewal through a payment vendor was discussed, financial information will not be not stored. Swami proposed PHP for the technology choice. Teruo Utsumi is working on a heavier duty system for membership management based on an open source package. The club's existing website runs on Wordpress (PHP) so the PHP will integrate well.

Selling of Club Assets

Dave Ittner opened discussion on the challenges of selling some of the more valuable gear which has been donated to the club. After discussion the approved process will be: 1)Make items available to members at a reasonable price, 2) If no member comes forward, attempt to sell at the swap meet, 3) If the item does not sell at the swap meet, sell via consignment to a dealer that the club is familiar with.

2017 Awards – Dave Ittner

Discussion of Awards for next years Annual Meeting – Dave has the forms and certificates for the awards and will upload them to the club Google Drive. We have a few awards in mind.

2/11/2017 Election

Rob Chapman presented action items for the 2017 Annual Election; there will likely be 3 board seats to fill. Rob C will notify membership of the open positions and encourage participation.

List of Club Assets

Dave Ittner proposed a project to create a list of club assets which could be maintained and easily transferred. Board member input for assets used by their respective programs was requested by the end of the year.

Ask SJAA

Dave Ittner began discussion on whether to continue the "Ask SJAA" service provided by the club with Lee Hoglan bowing out Spring 2017 after many years of service. Questions on how to continue with the program or whether to cancel it were discussed. Several ideas were presented but action pushed to the next board meeting.

Speaker System

Vini Carter provided an update on progress towards purchasing a new club speaker system. A $1000 budget was proposed by Vini and approved by all.

December 4, 2016

In attendance

Dave Ittner, Ed Wong, Rob Jaworski, Bill O'Neil, Vini Carter, Glenn Newell, Teruo Utsumi.
Absent: Rob Chapman, Lee Hoglan.
Guests: Tracy, Gary Chock, Tom Piller, Marianne, Amanda & Tyler

Ask SJAA

Dave Ittner asked the board members to monitor the inbox and respond to "Ask SJAA" requests if possible.

Club Correspondence

Dave Ittner asked board members to be more proactive in responding to club correspondence requests. Areas requiring more attention are: Program Updates, Ephemeris Review, Board Meeting minute review, and calls for Agenda Topics.

SkyMaps donation

Dave will research if donations are accepted by SkyMaps.com and if so notify Rob Jaworski who will make arrangements for a donation.

January 14, 2017

In attendance

Dave Ittner, Ed Wong, Rob Jaworski, Bill O'Neil, Vini Carter, Glenn Newell, Teruo Utsumi, Rob Chapman
Absent: Lee Hoglan; excused
Guests: Swami, George, Bruce Bastoky, Wolf Witt

Unitron Event

Dave Ittner notified the group that Bruce Bastoky has created a Unitron enthusiasts group and is planning a group weekend May 19-21, 2017. The group will be invited to the Friday ITSP and Dave has booked Houge for the club on 5/20 for their equipment presentations. Dave motioned the approval of SJAA for the Unitron Event and all approved.

2/11/2017 Elections

Ballots in progress; Rob Chapman will bring to the February meeting for voting. There are five director positions open for 2017. A new director, Wolf Witt, was appointed by the Board to fill Dave Ittner's vacated director position. Wolf will be up for re-election in 2018 along with 3 other board positions.

2/11/2017 Potluck

Teruo Utsumi created a Potluck signup sheet. The link is posted on Meetup.com event: www.perfectpotluck.com.

2/11/2017 Award Ceremony

Dave Ittner and Rob Jaworski reported that the selection of award recipients is in progress - Dave and Rob J will work out the details.

ASP Girl Scouts Pilot Program

Dave Ittner reported that the Astronomical Society of the Pacific is offering a training program on educating girls in STEM. Bill O'Neil will take the lead on finding interested members and picking a night for the training.

Solar Program - Batteries

Bill O'Neil expressed a need for new 35 AH Batteries for the solar program. Glenn Newell offered to put together a battery kit for the solar program. Bill O'Neil proposed a $300 budget for the batteries which was approved by all.

8/21/2017 Solar Eclipse

Bill O'Neil noted that the eclipse is coming soon and SJAA needs to begin planning for the event. Travel to the path of totality was discussed but will most likely not be pursued by SJAA as a club. Discussion was mostly about hosting an event locally to observe the partial eclipse. Teruo Utsumi will look for a candidate to drive an SJAA eclipse event program.

Club Calendar Operation

Dave Ittner reported that SJAA has several club calendars to help organize events and operations. Dave and Rob Chapman will work towards giving the board and officers the ability to view and edit the common club calendars and post directions to board@sjaa.net.

SJAA Library

SJAA offers another wonderful resource; a library with good astronomy books and DVDs available to all of our members that will interest all age groups and especially young children who are budding astronomers!
Please check out our wish list on the SJAA webpage:
http://www.sjaa.net/sjaa-library/

Telescope Fix It Session

Fix It Day, sometimes called the Telescope Tune Up or the Telescope Fix It program is a real simple service the SJAA offers to members of the community for free, though it's priceless. The Fix It session provides a place for people to come with their telescope or other astronomy gear problems and have them looked at, such as broken scopes whose owners need advice, or need help with collimating a telescope.
http://www.sjaa.net/programs/fixit/

Solar Observing

Solar observing sessions, headed up by Bill O'Neil, are usually held the 1st Sunday of every Month from 2pm - 4pm at Houge Park weather permitting. Please check SJ Astronomy Meetup for schedule details as the event time / location is subject to change:

http://www.meetup.com/SJ-Astronomy/

Quick STARt Program

The Quick STARt Program, headed up by Dave Ittner, helps to ease folks into amateur astronomy. You have to admit, astronomy can look exciting from the outside, but once you scratch the surface, it can get seemingly complex in a hurry. But it doesn't have to be that way if there's someone to guide you and answer all your seemingly basic questions.

The Quick Start sessions are generally held every other month.

http://www.sjaa.net/programs/quick-start/

Intro to the Night Sky

The Intro to the Night Sky session takes place monthly, in conjunction with first quarter moon and In Town Star Parties at Houge Park. This is a regular, monthly session, each with a similar format, with only the content changing to reflect what's currently in the night sky. After the session, the attendees will go outside for a guided, green laser tour of the sky, along with a club telescope to get a better look at celestial objects.

http://www.sjaa.net/programs/beginners-astronomy/

Loaner Program

Muditha Kanchana (Kanch) heads up this program. The Program goal is for SJAA members to be able to evaluate equipment they are considering purchasing or are just curious about by checking out loaners from SJAA's growing list of equipment. Please note that certain items have restrictions or special conditions that must be met.
If you are an SJAA member and an experienced observer or have been through the SJAA Quick STARt program please fill this form to request a particular item. Please also consider donating unused equipment.
http://www.sjaa.net/programs/loaner-telescope-program/

Astro Imaging Special Interest Group (SIG)

(SIG), February 2016, Bruce Braunstein agreed to lead the SIG group. SIG has a mission of bringing together people who have an interest in astronomy imaging, or put more simply, taking pictures of the night sky. The Imaging SIG meets roughly every month at Houge Park to discuss topics about imaging. The SIG is open to people with absolutely no experience but want to learn what it's all about, but experienced imagers are also more than welcome, indeed, encouraged to participate. The best way to get involved is to review the postings on the SJAA Astro Imaging mail list in Google Groups.

http://www.sjaa.net/programs/imaging-sig/

Astro Imaging Workshops & Field Clinics

Not to be confused with the SIG group this newly organized program championed by Glenn Newell is a hands on program for club members, who are interested in astro-photography, to have a chance of seeing what it is all about. Workshops are held at Houge Park once per month and field clinics (members only) once per quarter at a dark sky site. Check the schedule and contact Glenn Newell if you are interested.

School Star Party

The San Jose Astronomical Association conducts evening observing sessions (commonly called "star parties") for schools in mid-Santa Clara County, generally from Sunnyvale to Fremont to Morgan Hill.

Contact SJAA's Jim Van Nuland (Program Coordinator) for additional information.

http://www.sjaa.net/programs/school-star-party/

SJAA Contacts

President:	Dave Ittner
Vice President:	Ed Wong
Treasurer/Dir:	Rob Jaworski
Secretary/Dir:	Rob Chapman
Director:	Bill O'Neil
Director:	Teruo Utsumi
Director:	Vini Carter
Director:	Glenn Newell
Director:	Wolf Witt
Director:	Swami Nigam
Director:	Sukhada Palav
Ephemeris Newsletter -	
Editor:	Sandy Mohan
Prod. Editor:	Tom Piller
Fix-it Program:	Vini Carter
Imaging SIG:	Bruce Braunstein
Intro to the Night Sky:	David Grover
Library:	Sukhada Palav
Loaner Program:	Muditha Kanchana
Memberships:	Anand Rajagopalan
Publicity:	Rob Jaworski
Questions:	Lee Hoglan
Quick STARt	Dave Ittner
Solar & Starry Nights:	Bill O'Neil
School Events:	Jim Van Nuland
Speakers:	Sukhada Palav

E-mails: http://www.sjaa.net/contact

SJAA Ephemeris, the newsletter of the San Jose Astronomical Association, is published quarterly.

Articles for publication should be submitted by not later than the 20th of the month of February, May, August and November. (earlier is better).

San Jose Astronomical Association
P.O. Box 28243
San Jose, CA 95159-8243
http://www.sjaa.net/contact

San Jose Astronomical Association
P.O. Box 28243
San Jose, CA 95159-8243

Place
postage
here

Fold here

San Jose Astronomical Association Annual Membership Form

P.O. Box 28243 San Jose, CA 95159-8243

Membership Type:

☐ **New** ☐ **Renewal** (Name only if no corrections)

☐ **$20** Regular Membership with **online** Ephemeris

☐ **$30** Regular Membership with **hardcopy** Ephemeris mailed to below address

The newsletter is always available online at: http://www.sjaa.net/sjaa-newsletter-ephemeris/

Questions? Send e-mail to: memberships@sjaa.net

Bring this form to any SJAA Meeting or send to the address (above). Make checks payable to "SJAA", or join/renew at: http://www.sjaa.net/join-the-sjaa/

Name: ______________________________

Address: ______________________________

City/ST/Zip: ______________________________

Phone:

E-mail address: